

**Time:** 5 minutes**Outcome:** Connect expected inflation to nominal and real interest rates.**Difficulty:**

Intermediate

Fisher Effect



THE CORE IDEA

The Fisher effect links the nominal interest rate to the real interest rate and expected inflation. The nominal rate approximately equals the real rate plus expected inflation, so lenders generally require a higher nominal rate when they expect faster price growth. This adjustment helps preserve the expected real return, though unexpected inflation can still redistribute purchasing power.



HOW TO RECOGNIZE IT

- The Fisher equation links nominal rates, real rates, and expected inflation: nominal rate approximately equals real rate plus expected inflation.
- Nominal interest rates include expected inflation, so they rise when expected inflation rises.
- The Fisher effect says higher expected inflation tends to raise nominal interest rates.
- Higher expected inflation tends to be built into higher nominal interest rates.
- A bank sets a fixed nominal loan rate today for next year.



WATCH OUT

Do not use realized inflation to explain a nominal rate chosen before the loan was made. The nominal rate written into a contract reflects expected inflation at the time the contract is set; realized inflation determines the ex post real return.



WORKED EXAMPLE: APPLYING THE FISHER EQUATION

Real interest is 2% and expected inflation is 6%. The usual approximation gives nominal interest of 2% + 6% = 8%. The exact compounding calculation is $(1.02 \times 1.06 - 1) \times 100 = 8.12\%$. At a given real rate, higher expected inflation raises the nominal rate lenders require.



CHECK YOURSELF

A borrower says the Fisher effect is found by subtracting this year's actual inflation from a loan rate set last year. What is the key mistake?

**READY?**

Return to the game and master this concept.